

Can a forward osmosis-reverse osmosis hybrid system achieve 90 % wastewater recovery and desalination energy below 1 kWh/m³? A design and simulation study

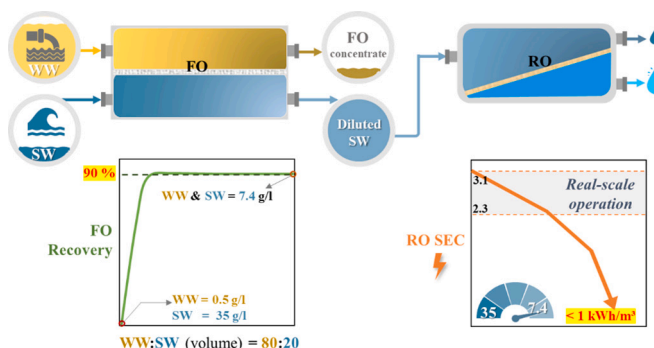
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HIGHLIGHTS

- WW served as feed (FS) and SW (35 g/l) as draw (DS) in FO-RO hybrid.
- Simulated FS salinities (0.25–1.5 g/l) and FS:DS volume ratios (50:50 to 90:10)
- At 0.5 g/l FS and 80:20 FS:DS (volume ratio), FO achieved 90 % FS recovery.
- Osmotic dilution in FO-RO hybrid allow for RO operation below 1 kWh/m³.
- Existing RO demonstrated adaptability to FO-RO integration.

GRAPHICAL ABSTRACT



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ABSTRACT

The emergence of the forward osmosis (FO)-reverse osmosis (RO) hybrid offers simultaneous water reuse and desalination by incorporating FO for wastewater (WW) recovery to dilute seawater (SW) before undergoing RO treatment. The potential of osmotic dilution within the FO-RO hybrid promises elevated FO recovery rates and decreased specific energy consumption (SEC) of RO. Yet, pushing the system to its limits remained unexplored. Employing MATLAB® for simulations across various feed-to-draw volume ratios and diverse WW salinities, the study explored the feasibility of achieving enhanced FO recovery from WW. The study extended to evaluating the adaptability of existing desalination plants to process diluted SW using LewaPlus® design software. The FO simulation confirmed the feasibility of achieving 90 % WW recovery across various operational conditions in many instances. Yet, the optimal configuration for achieving both 90 % WW recovery and RO SEC below 1 kWh/m³ was specific: an 80:20 feed-to-draw volume ratio with an initial WW salinity of 0.5 g/l in FO. This resulted in reduced SW salinity (7.4 g/l), enabling RO desalination at 0.96 kWh/m³, showcasing potential retrofitting applications for existing desalination plants based on software design response. In the comparative study to determine the optimal approach for integrated water reuse and desalination between FO-RO hybrid and independent schemes, the energy efficiency of RO in independent schemes appears favorable, but FO-RO hybrid system may offer other benefits considering its double-dense barrier protection.

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Nomenclature		SWRO	seawater reverse osmosis
Acronyms AL-FS active layer facing feed solution CCRO closed-circuit reverse osmosis CTA cellulose triacetate DS draw solution ECP external concentration polarization ED electrodialysis ERD energy recovery device FO forward osmosis FS feed solution HF hollow fiber ICP internal concentration polarization MED multi-effect distillation MF micro-filtration MSF multi-staged flash MSRO multi-staged reverse osmosis PV pressure vessel R recovery RO reverse osmosis RSD reverse salt diffusion SCRO semi-closed reverse osmosis SW seawater		TFC	thin film composite
		UF	ultra-filtration
		VC	vapor compression
		WW	wastewater
		Symbols	
		$\pi_{D \text{ bulk}}$	bulk osmotic pressure in draw (bar)
		$\pi_{F \text{ bulk}}$	bulk osmotic pressure in feed (bar)
		π_D	osmotic pressure in draw (bar)
		π_F	osmotic pressure in feed (bar)
		A	membrane-specific pure water permeability (lmh/bar)
		B	membrane-specific solute permeability (m/s)
		C	centigrade
		D	self-diffusion coefficient of solute (m^2/s)
		h	hour
		J_s	reverse solute flux ($\text{g}/\text{m}^2.\text{h}$)
		J_w	water flux (lmh)
		K	resistance to solute diffusion (s/m)
		k	solute mass transfer coefficient (m/s)
		kW	kilowatt
		S	structural parameter (mm)
		SEC	specific energy consumption (kWh/m^3)
		Els	no. of membrane elements

1. Introduction

Shifting climate patterns, a burgeoning global population with rising food requirements, economic expansion, and the scarcity of freshwater resources have compelled the water industry to refocus its efforts on incorporating unconventional water sources into sustainable, long-term water production strategies. In this context, seawater (SW) desalination, known for its capacity to augment water resources independent of natural hydrological cycles, and the reuse of reclaimed water, offer a reliable water supply [1,2]. Desalination technologies historically relied on thermal methods like multi-staged flash (MSF), multi-effect distillation (MED), and vapor compression (VC), known for their relatively high specific energy consumption (SEC) ranging from 13 to 22 kWh/m^3 [3]. However, with the advent of reverse osmosis (RO), the desalination landscape underwent a revolution, achieving significantly lower SEC values of 7–8 kWh/m^3 . Technological advancements, such as integrating energy recovery devices (ERDs) like Francis turbines in the early 1990s, further reduced SEC to 4–5 kWh/m^3 . Subsequent innovations, including the adoption of Pelton wheel, further decreased SEC to 3–4 kWh/m^3 . The implementation of isobaric Energy Recovery Devices (ERDs) then brought SEC to less than 3 kWh/m^3 , signifying a clear progression towards increasingly efficient energy consumption in desalination processes [4]. RO is considered as the most optimized membrane-based desalination process, making it the current dominant and reliable state-of-the-art technique for desalting SW compared among other desalination techniques like MSF, MED, membrane distillation (MD), and electrodialysis (ED) [5,6]. Despite of advancements in development of thin film composite (TFC) RO membranes with highly permeable and ERDs in recent decades [7], there still remains a substantial room to further reduce energy demand.

The total specific energy consumption (SEC) for a typical real-scale one-pass seawater reverse osmosis (SWRO) plant with single-stage, equipped with ERD and operated at 45 % recovery can range from 3.8 to 4.5 kWh/m^3 [8], depending on operational variables such as feed conditions (salinity and temperature), target product quality, ERD type, and membrane unit configuration. Notably, within this total energy consumption range, approximately 1 kWh/m^3 [9,38], is allocated for pre- and post-treatment operations, regardless of diverse operational

conditions. Post-treatment, encompassing the operation of the second-pass RO, is vital for meeting strict standards regarding boron and chloride concentrations in desalinated water. This is indispensable for applications such as agricultural irrigation in water-scarce regions, drinking water, and industrial use, with an associated energy consumption of 0.5 kWh/m^3 [10]. Certainly, the substantial energy allocation, ranging from 2.3 to 3.1 kWh/m^3 [8], features the RO system's role as the most energy-intensive component within the desalination plant. This underscores the importance of ongoing research efforts aimed at reducing this energy demand, highlighting it as one of the most actively studied areas in the desalination industry.

The thermodynamic minimum specific energy required for desalinating a 35 g/l feed with 50 % recovery was determined to be 1.06 kWh/m^3 [9]. Practical operation demands a considerable additional energy input, emphasizing the need for advanced configurations to further enhance the efficiency of RO. The viability of RO in addressing water and energy nexus lies in innovations that reduce its energy demand.

In this context, exploration of alternative RO configurations that minimize applied pressure are of particular interest. Theoretically, a multi-staged reverse osmosis (MSRO) design could approach the minimum specific energy requirement by pressurizing smaller feed volumes in each stage, practical challenges impede its implementation. Even in two-stage systems, potential energy savings might not justify the added capital costs, involving costly components like booster pumps, extra pressure vessels (PV), and additional auxiliary equipment, along with more RO membrane modules [7,9]. Despite the clear advantage of reducing specific energy, the substantial increase in capital expenses has hindered the broad adoption of multi-stage RO systems. Closed-circuit reverse osmosis (CCRO) achieves energy savings by cyclically reintroducing concentrate to the feed, allowing a proportional increase in applied pressure. Yet, mixing in CCRO may elevate entropy and energy use, especially at higher recovery rates, requiring scrutiny for long-term performance [11,12]. Furthermore, CCRO struggles to entirely eliminate scaling, with membrane fouling persisting, notably in the presence of organic and biofouling alongside mineral scaling [13]. Semi-closed reverse osmosis (SCRO) process optimizes energy efficiency by avoiding feed-concentrate mixing, employing separate cycles with pressurization above osmotic pressure and consumes an ideal energy of 1.14

kWh/m³ at 50 % recovery. However, it still requires capital expenditure and further validation through pilot and large-scale trials [14].

Despite advancements in RO technology, achieving energy efficiency close to the thermodynamic limit remains challenging due to high capital costs and the necessity for careful evaluation to overcome drawbacks in each, impeding the widespread adoption of these technologies. Recognizing these challenges, an alternative approach involves osmotic dilution, strategically targeting a reduction in feed salinity rather than attempting to bridge the gap to the thermodynamic limits in SWRO.

Operating parameters such as temperature, pressure, and feed water salinity play a crucial role in influencing the efficiency of the RO process [15]. The energy consumption in RO is affected by the salinity of incoming saline water, exhibiting a nearly linear increase with applied pressure [9]. The hydraulic pressure required to counteract the osmotic pressure in the feed is greater when dealing with full-strength SW as opposed to diluted SW, which translates into more economical desalination process [16]. Forward Osmosis (FO) can efficiently extract water using a saline stream which has been envisioned for SW desalination [17]. Forward osmosis (FO) come in two operational configurations: open-loop and closed-loop. In an open-loop system (Fig. 1a), the draw solution (DS) is used once without recirculation or regeneration, contrasting with the closed-loop system (Fig. 1b). However, the DS dilution step in FO within closed-loop FO-RO hybrid poses an inherent energy challenge during DS regeneration, falling short of outperforming traditional desalination methods [18].

The process of osmotic dilution using FO has the potential to lower the RO feed salinity, offering an integrated approach to SW desalination and water reuse, commonly known as the FO-RO hybrid system [19]. Resulting from solute concentration difference, water from feed solution (FS) (impaired water source i.e., wastewater (WW) effluent) naturally diffuses through the FO membrane into DS (SW stream) which later undergoes subsequent desalination in RO. Hence the integration of simultaneous water reuse and desalination is especially significant for open-loop FO-RO hybrid system. In FO, the SEC is limited to circulating FS and DS, typically ranging between 0.09 and 0.55 kWh/m [20,21,33]. Beyond energy savings attributed to feed dilution in RO process, this controlled predilution offers reduced RO membrane fouling, double-barrier protection, beneficial water reuse and lowers the RO brine concentration [17]. This integrated approach may eliminate the necessity of a two-pass RO, unlike conventional desalination plants that cannot achieve low boron and chloride concentrations with a single pass, even when employing high-rejection membranes [10].

Many studies demonstrated using FO with WW as FS for osmotic dilution prior to SWRO showed potential in reducing RO's SEC and optimized operation. Yangali-Quintanilla et al. showed that SW osmotic dilution uses 27 % less energy than traditional desalination offering a payback period of less than 1.5 years [22]. Teusner et al. found FO-RO hybrid to be 12 % cheaper and 7 % more energy-efficient than conventional desalination, recommending a FO flux of 15 lmh for optimal

cost-effectiveness [23]. Life cycle cost assessment conducted by Valladares et al. stated that FO-LPRO configurations exhibit a 21 % higher capital expenditure but offer a significant 56 % reduction in operational expenditure due to substantial energy consumption savings [24]. Sao et al. found that a hybrid FO-RO system, when used for simultaneous desalination and WW reclamation, offers cost savings of 24.7 % compared to standalone RO systems [25]. In their thorough investigation of FO's utility as an osmotic dilution and concentration step in practical applications, Zhan et al. found the energy consumption of FO-RO hybrid systems is markedly lower compared to conventional technologies like RO, when employed for direct and indirect SW desalination and WW reclamation applications [26]. Within the exploration of prior research on pilot-scale FO-RO hybrid systems, Nassrullah et al. found that, besides the energy benefits of FO-RO hybrid systems, the RO component achieved higher recovery rates compared to standalone RO systems [3]. In their exploration of FO-RO hybrid technology for treating WW from a coal-fired power plant and SW as draw, Gyu Choi et al. observed a substantial reduction in RO fouling behavior, accompanied by a significant 15 % decrease in the energy consumption of the FO-RO osmotic dilution process compared to a conventional SWRO process [20]. In their broader analysis of prior research on FO-RO hybrid systems, they identified a total energy consumption range of 1.5 to 3.49 kWh/m³ for WW treatment and desalination in an FO-RO hybrid system.

As RO approaches its thermodynamic limit, the energy and cost savings from adding FO are expected to outweigh the investment [17,27]. Therefore, it is crucial to consider the optimal advantage of potential osmotic dilution for achieving maximum energy savings in RO. Reported energy savings in modeling [10,19,28–33], bench-scale [34] and pilot-scale [20,33] studies on FO-RO hybrid systems are capped to a maximum FO recovery of 55 %, representing the current state-of-art. This falls well below the expected recovery of WW in water reuse schemes (typically 80–90 %) [35] underscoring the need for further investigation. Within the reported energy consumption range of 1.17 to 1.4 kWh/m³ [35] for potable water reuse, exploring FO's potential to achieve 90 % WW recovery stands as a pivotal strategy. This robust approach ensures maximal benefits in RO energy, making FO-RO integration imperative for optimizing integrated water reuse and desalination approach. It is noteworthy that a subset [20,31,34] of these studies did not account for RO energy with ERD considerations. Thus, the full potential of harnessing the synergy between FO-RO hybrid for further reducing RO energy needs remained untapped.

Expanding on these insights, current study aimed to evaluate the feasibility of the FO-RO hybrid system through a design study. The focus is twofold: achieving a robust WW recovery rate of 90 % through FO and evaluate if simultaneously desalinating the diluted SW with RO SEC of 1 kWh/m³ or less is achievable. At first, a systematic study was performed to define key parameters of FO to achieve 90 % recovery considering WW and SW streams salinities. Then based on the resulting diluted streams from FO, SEC of RO, and feasibility of retrofitting existing RO plant was assessed. Additionally, a comparative study on SEC of RO in

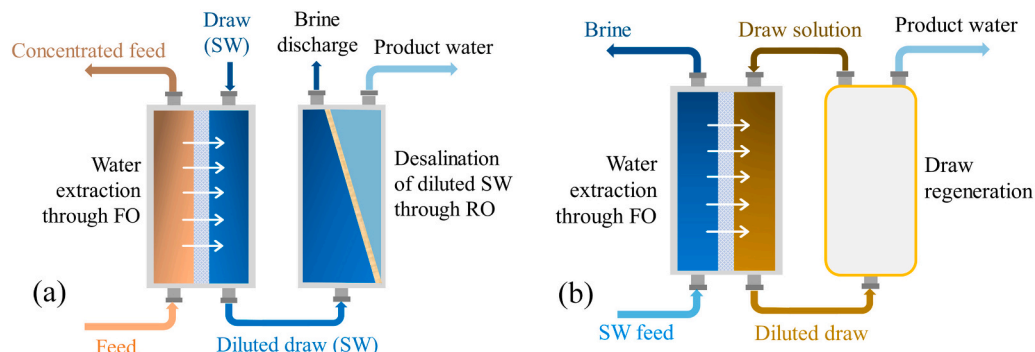


Fig. 1. Operational configurations of FO-RO hybrid system (a) open-loop, (b) closed-loop.

the integration of water reuse and desalination through independent schemes (i.e., separate water reuse and desalination schemes) and FO-RO hybrid were presented to assess the energy efficiency of each approach.

2. Materials and methods

2.1. MATLAB-based modeling and simulation

The FO semipermeable membrane harnesses osmotic pressure gradients within solutions to facilitate separation of solvent molecules from diluted FS side to the concentrated DS side. Taking into account mass transfer limitations, such as internal concentration polarization (ICP) that takes place within the support layer where solvent permeates from the feed solution, resulting in dilution of the draw concentration within the active layer, FO also faces external concentration polarization (ECP). ECP occurs on both the feed and draw sides, disrupting the effective osmotic pressure balance, with concentrative and dilutive concentration polarization affecting each side, respectively. ECP can be mitigated by optimizing hydrodynamic conditions like elevated crossflow velocity at the surface of membrane [36]. Besides ICP and ECP, reverse salt diffusion (RSD) represents another facet of mass transport theory in FO. An ideal membrane should reject the transfer of dissolved solutes from the DS to the FS, some solute particles might diffuse through the membrane, leading to RSD. Considering the effects of ICP, ECP and RSD on FO flux, an advanced solution diffusion model for active layer facing FS (AL-FS), expressed by a highly non-linear equation [19] was modelled in co-current mode, and solved in MATLAB® (R2022b), denoted in Eq. (1).

$$J_w = A \cdot \left[\frac{\pi_{D \text{ bulk}} \cdot e^{-J_w \cdot K} - \pi_{F \text{ bulk}} \cdot e^{\frac{J_w}{k}}}{1 - \frac{B}{J_w} \cdot (e^{-J_w \cdot K} - e^{\frac{J_w}{k}})} \right] \quad (1)$$

In Eq. (1), J_w : water flux, A: membrane-specific pure water permeability, B: membrane-specific solute permeability, k: solute mass transfer coefficient, K: resistance to solute diffusion, $\pi_{D \text{ bulk}}$: osmotic pressure in draw, $\pi_{F \text{ bulk}}$: osmotic pressure in feed. The reverse solute flux (J_s) influencing the FO flux is calculated using Eq. (2) [36].

$$J_s = B \cdot \left[\frac{C_{D \text{ bulk}} \cdot e^{-J_w \cdot K} - C_{F \text{ bulk}} \cdot e^{\frac{J_w}{k}}}{1 - \frac{B}{J_w} \cdot (e^{-J_w \cdot K} - e^{\frac{J_w}{k}})} \right] \quad (2)$$

In Eq. (2), $C_{D \text{ bulk}}$: bulk draw concentration, $C_{F \text{ bulk}}$: bulk feed concentration.

Fouling, even if it can present a significant constraint on FO performance affecting the flux and therefore membrane surface area required, was not considered in this initial study since it does not directly

influence the osmotic equilibrium.

Initially, the model underwent a numerical simulation study with a 50:50 initial feed-to-draw volume ratio. The initial salinity of the FS was 0.5 g/l, while the DS had an initial salinity of 35 g/l. [27] In the simulation, iterative steps were taken to systematically increase the membrane area per square meter until the system reached osmotic equilibrium. Fig. 2 illustrates the input parameters, the numerical simulation process and the resulting outcomes using MATLAB.

This exploration was aimed to comprehend the impact of membrane properties on critical parameters, including water flux, final recovery, and the requisite surface area for achieving osmotic equilibrium across a range of commercially available membranes. Detailed characteristics of the membranes considered in the study are outlined in Table 1.

Then a thorough analysis was conducted to assess the impact of varying feed-to-draw ratios on the potential of FO recovery for diluting the DS, considering membrane characteristics. The study included variations in feed-to-draw volume ratios, namely, 50:50, 60:40, 70:30, 75:25, 80:20, and 90:10, while maintaining a constant initial FS salinity of 0.5 g/l. Furthermore, the influence of varying FS initial salinity on FO recovery, considering values of 0.25, 0.5, 0.75, 1.0, 1.2, and 1.5 g/l, with a constant feed-to-draw volume ratio of 50:50 was performed. The study utilized WW with a salinity range of 0.25–1.5 g/l, originating from secondary treated WW sources, reflecting its low saline nature [40,41]. This analysis provided insights into the implication of diverse operating conditions in reaching the osmotic equilibrium and corresponding effect on FS recovery and DS dilution.

To enhance the optimization of WW recovery and SW intake, it is crucial to gain a comprehensive understanding of FO performance across a diverse range of FS salinities and feed-to-draw volume ratios. A comprehensive examination of the model's response under various operating conditions was conducted to elucidate the FO performance, focusing on the achievable final FS recovery and diluted DS salinity.

Table 1

Characteristics of commercially available membranes considered in the study.

Manufacturer/material/ configuration	A (lmh/ bar)	B (m/s)	S (mm)	Reference
HTI/cellulose triacetate (CTA)/ flat sheet	0.7	1.50E-07	0.663	[37]
HTI/TFC/flat sheet	1.3	3.00E-08	1.227	
Porifera/TFC/flat sheet	2.0	1.20E-07	0.344	
Aquaporin/TFC/hollow fiber	1.56	6.67E-08	0.430 ^a	[38]
Aquaporin/TFC/flat sheet	0.52	2.50E-08	0.825	[39]
Toray/TFC/flat sheet	5.362	2.64E-07	0.266	
Toyobo/CTA/hollow fiber	0.545	1.11E-08	0.639	

^a In the model, the structural parameter (S) was adjusted to align with the flux values reported in the literature.

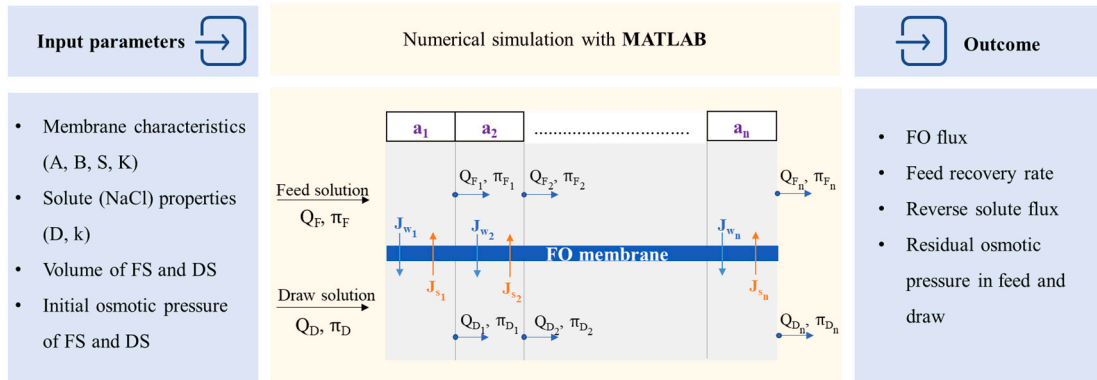


Fig. 2. FO process simulation followed in MATLAB. a: unit area of membrane; Q_F and Q_D : FS and DS volume respectively; π_F , π_D : FS and DS osmotic pressure; J_w : water flux; J_s : reverse solute flux.

2.2. RO stage design

Building on the FO performance analysis, the study advanced to evaluate the SEC of the RO process required to desalinate the diluted SW. The utilization of LANSXESS's LewaPlus® software streamlined the design and performance analysis of the RO system. For a comprehensive evaluation, an established desalination plant initially designed for desalination of full-strength SW underwent adaptation to process osmotically diluted SW. This strategic retrofitting allowed us to gauge the adaptability of existing SWRO systems for FO-RO hybrid applications. The approach facilitated a thorough examination, providing insightful data on the energy dynamics of RO in the integrated FO-RO system. The detailed characteristics existing desalination plant were presented in Table 2 [19]. The study considered initial baseline assumptions for RO feed salinities in the diluted SW (resulting from FO), ranging from 20 to 5 g/l. Fig. 3 illustrates the proportion of FO permeate from WW and SW to achieve the specified salinities under consideration.

Furthermore, following scenarios were simulated to comprehend the optimized RO performance for diluted SW feed. (i) Increased RO recovery resulting from diluted feed with the same baseline design, and (ii) considering a new RO system where LewaPlus® chooses the desired number of PVs and membrane elements, aiming to achieve the same enhanced recovery at respective RO feed salinities. Subsequently, a comparative analysis was performed aimed to determine the optimal approach between integrated water reuse and desalination within an FO-RO hybrid system and independent schemes.

3. Results and discussion

3.1. MATLAB numerical simulation results on FO performance

In the preliminary outcomes of the simulation study, various commercial membranes were evaluated under a 50:50 feed-to-draw ratio and 0.5 g/l initial FS salinity conditions. The study explored surface area requirements for reaching osmotic equilibrium, (Fig. 4a) and the maximum achievable recovery, (Fig. 4b). The FO recovery rates in this study were determined through a mass transport model, exemplify advancements that surpass previously reported recovery rates [19,20,28–34].

All the membranes allowed to reach 90 % recovery of FS. After reaching this targeted recovery, all the membranes efficiently reached osmotic equilibrium without the need for extensive surface area. Notably, membranes like Toray, Porifera, and Aquaporin HF achieved early osmotic equilibrium with relatively less surface area, resulting in recoveries of 95.98 %, 95.72 %, and 96.13 %, respectively. In contrast, Aquaporin flat sheet, HTI-CTA, HTI-TFC, and Toyobo, requiring twice the surface area to reach osmotic equilibrium, still achieved notable recoveries of 96.01 %, 92.15 %, 96.61 %, and 96.68 %, respectively.

This observed difference in the surface area required to reach osmotic equilibrium was attributed to the influence of membrane characteristics. Ideal membrane should allow for high permselectivity, i.e., high water flux and low salt passage. In the context of FO, concentration polarization (especially ECP) is a major limitation for osmotic pressure

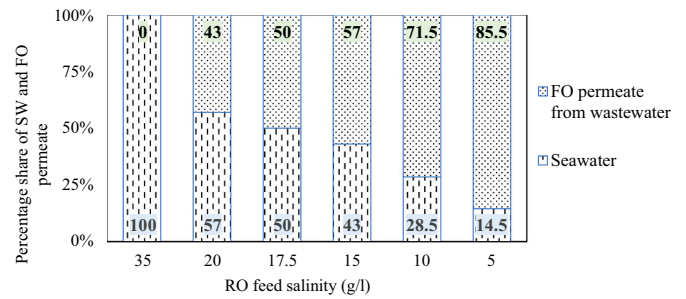


Fig. 3. Composition of SW and FO permeate from wastewater blend for different RO feed salinities.

efficiency. As such, ideal membrane will feature high pure water permeability coefficient (A), low solute permeability coefficient (B), and small structural parameter (S), achieve osmotic equilibrium earlier with relatively less surface area compared to membranes with different properties [42,43]. For instance, the Toray membrane, characterized by higher A, reaches osmotic equilibrium early with less surface area compared to other membranes. Despite its high B value, which greatly influences reverse solute diffusion across the membrane, the permeability-selectivity trade-off mitigates the need for additional surface area to reach osmotic equilibrium, influenced by its low S. Although Toray, Porifera, and Aquaporin HF achieve early osmotic equilibrium with relatively less surface area, there still exists minimal difference in their maximum achievable recovery rates, dictated by their permeability-selectivity trade-off. Conversely, HTI-CTA, HTI-TFC, and Toyobo, with relatively low A values, require more surface area to reach osmotic equilibrium but still deliver high recoveries, determined by their low B and S values. This observed diversity in performance has profound implications, particularly in terms of the economic considerations associated with membrane surface area requirements for full-scale applications. The cost implications of selecting membranes with varying performance characteristics should be thoroughly evaluated to optimize the overall efficiency and economic viability of the FO process.

Given that all commercial membranes met the 90 % recovery target, Aquaporin HF was chosen for further study on various operating conditions due to its ability to attain 96 % FS recovery and osmotic equilibrium using minimal surface area. In subsequent simulations involving diverse operating conditions, including variations in feed-to-draw volume ratios and different initial FS salinities, the Aquaporin HF membrane exhibited notable trends. When decreasing the DS volume from 50:50 to 90:10, while maintaining an initial FS salinity of 0.5 g/l (Fig. 5a), the membrane required an increasing surface area to reach osmotic equilibrium, coinciding with a significant decline in the maximum achievable FS recovery, (Fig. 5b). Similarly, when varying FS salinity from 0.25 g/l to 1.5 g/l and at a constant feed-to-draw volume ratio of 50:50, the membrane exhibited an increased surface area requirement to achieve osmotic equilibrium, (Fig. 5c), accompanied by a decline in FS recovery with an increase in FS salinity, (Fig. 5d). It is noteworthy that the diminishing osmotic pressure gradient during the optimization of SW intake had a more pronounced effect on FS recovery compared to variations in FS salinity, leading to higher recoveries in the latter case.

This highlights the sensitivity of the FO process to variations in SW intake and FS salinity, underscoring the importance of careful optimization for achieving desired recovery rates. Comprehensive understanding of FO performance across diverse FS salinities and volume ratios is essential for optimizing WW recovery and SW intake. The achievable final FS recovery and final DS salinity across varying initial FS salinity and feed-to-draw volume ratios were presented in Fig. 6(a) and (b) respectively.

Fig. 6(a) illustrates the feasibility of achieving robust FS recovery rates of 90 % and above within the diverse operating conditions

Table 2

Key parameters of existing RO desalination plant undergoing retrofit for osmotically diluted SW operation.

Source water	SW (35 g/l)
Temperature	20 °C
Plant capacity	1800 m ³ /h
Total recovery (R)	45 %
Configuration	1 pass
Membrane element	SW HR 400 34
No. of PV	477
No. of elements (Els)/PV	7
ERD	Isobaric ERD

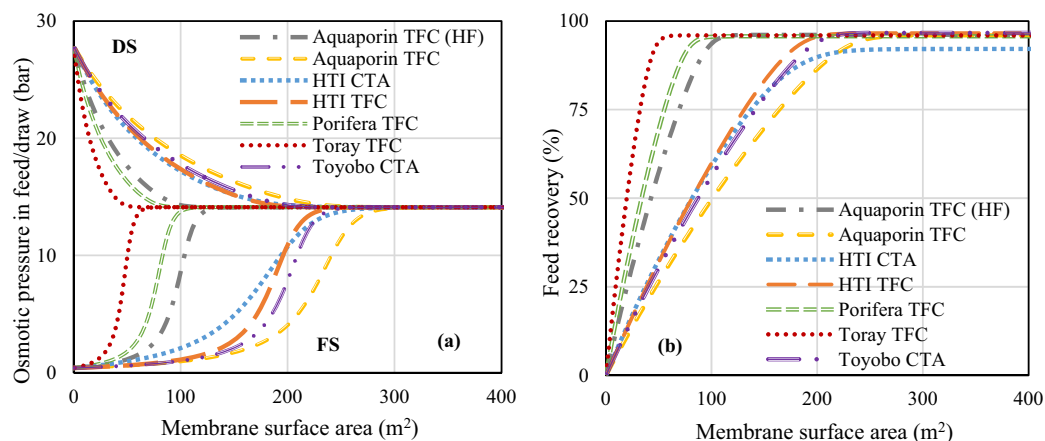


Fig. 4. Comparative performance analysis of commercially available membranes when subjected to 50:50 feed-to-draw volume ratio with initial FS salinity at 0.5 g/l and DS salinity 35 g/l. (a) Evolution of osmotic pressure equilibrium, (b) achievable FS recovery.

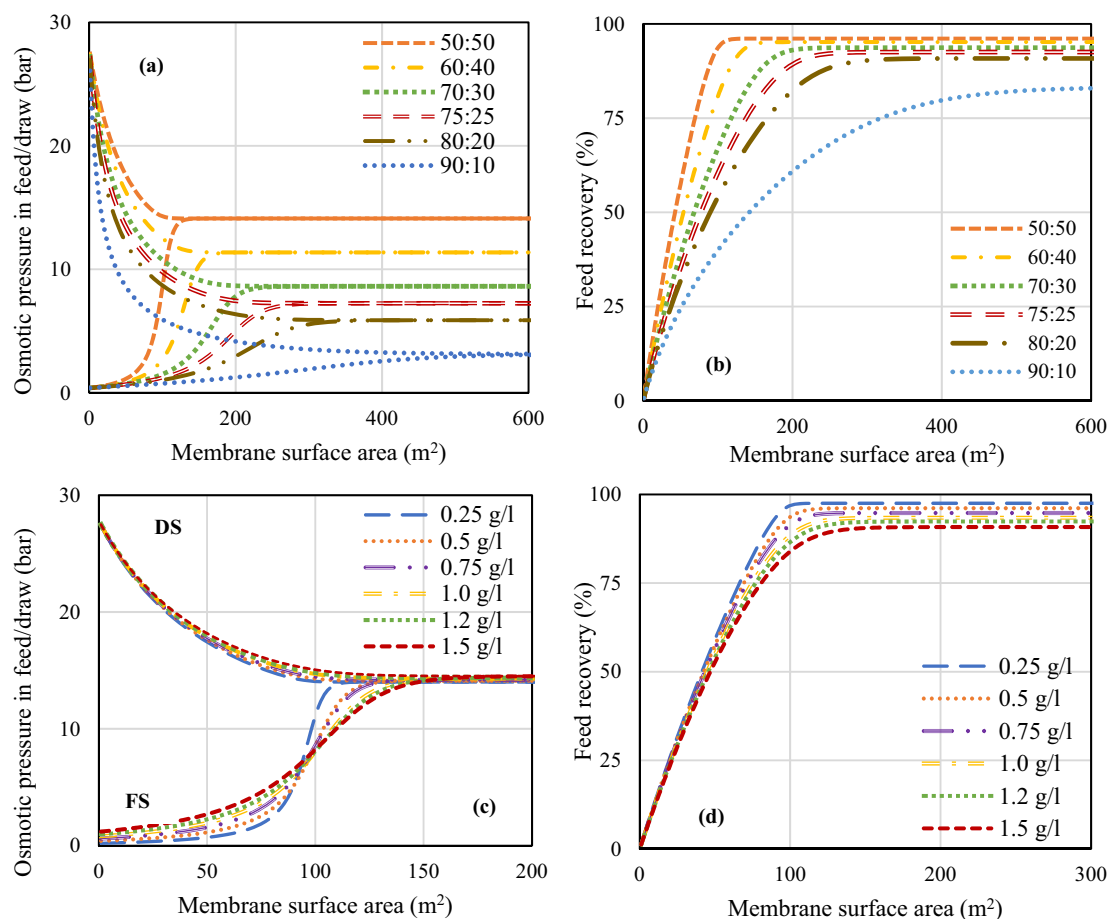


Fig. 5. Performance analysis of Aquaporin HF membrane under diverse operating conditions. (a) Evolution of osmotic pressure equilibrium and (b) achievable recovery rate in response to varying feed-to-draw volume ratio and 0.5 g/l of initial FS salinity; (c) evolution of osmotic pressure equilibrium and (d) achievable recovery rate in response to varying initial FS salinity and constant feed-to-draw ratio of 50:50.

considered. For initial FS salinities of 0.25 and 0.5 g/l, FO demonstrated recovery rates surpassing 90 % for feed-to-draw volume ratios up to 80:20. However, as both FS salinity and volume increased, FO experienced a decline in FS recovery, falling short of maintaining a 90 % level in all instances. In essence, the recovery rates of FS at 90 % and above were more prominently observed, especially for low initial FS salinities, extending to high feed-to-draw ratios. Conversely, with an increase in FS salinity, the recovery rates at 90 % and above became more pronounced

for lower feed-to-draw volume ratios. Notably, at feed-to-draw volume ratios from 50:50 to 70:30 and across all considered initial FS salinities, the recovered FS resulted in SW dilution levels exceeding 10 g/l (Fig. 6b). The achievement of both 90 % water recovery and a decrease in DS salinity to below 10 g/l was realized by maintaining an optimal FS volume within the 75 % to 80 % range, combined with FS salinities of 0.25 and 0.5 g/l.

Given the outcome of the FO study with an optimized ratio of WW:

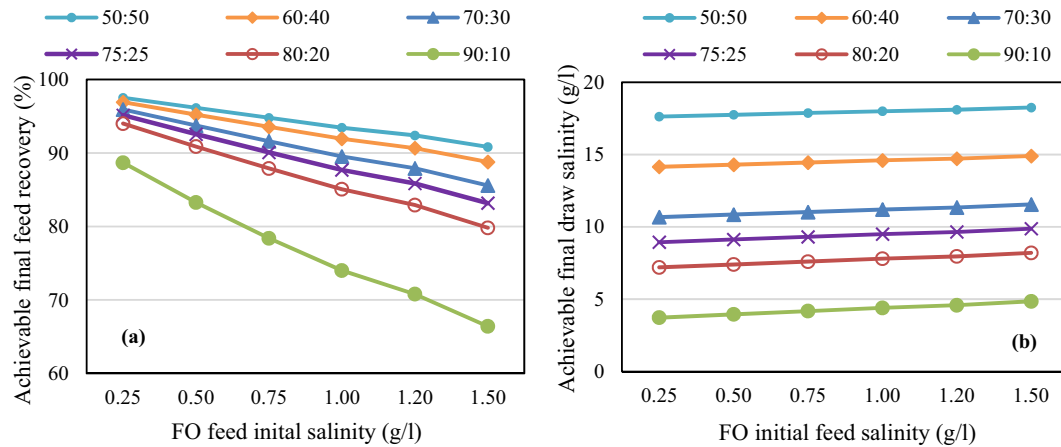


Fig. 6. Analyzing Aquaporin HF membrane performance w.r.t varying feed-to-draw volume ratios and varying initial FS salinities. (a) Achievable final percentage of FS recovery; (b) achievable final diluted DS salinity.

SW of 80:20 to achieve high DS dilution and a targeted 90 % recovery of the WW, important constraints arise regarding the FO design to assure good hydrodynamics all along the process [44,45]. Apart from the evaluation of operation in counter-current mode, further work and engineering study may consider optimized module design and their arrangement and stages to assure adequate operation.

3.2. Enhanced RO energy efficiency

Fig. 7(a) depicts the SEC of RO necessary for desalting the diluted SW, with a consistent plant recovery of 45 % to yield a permeate of 1800 m³/h. LewaPlus® identified no design warnings or errors, affirming the feasibility of this retrofitting. Transitioning to Fig. 7(b), it delves into the realm of energy efficiency presenting a comparative analysis of the percentage energy savings resulting from osmotically diluted RO feed, providing a direct comparison between the energy consumption of the proposed system and traditional one-pass and two-pass RO configurations. In all the cases, ERDs, aiming to minimize energy consumption by reclaiming hydraulic pressure from a pressurized brine stream were implemented [46].

Thanks to osmotic dilution, SEC of RO stage showed a proportional decrease as feed salinity decreases. Notably, the same quantity of permeate can be produced with half the energy i.e., 1.05 kWh/m³ compared to a one-pass RO system when the RO feed salinity was 10 g/l. Remarkably, at a feed salinity of 5 g/l, the SEC of the RO system drops

even further, registering below 1 kWh/m³. This study, uniquely, explored the possibilities of reducing RO SEC below 1 kWh/m³ through FO-RO hybrid, a dimension not previously addressed in existing literature [19,20,28–34].

3.3. Optimizing configuration for 90 % WW recovery and achieving below 1 kWh/m³ SEC of RO in FO-RO hybrid

The FO simulation results provided insights into the achievable diluted SW salinities under various operating conditions. Fig. 8 illustrates the RO SEC needed to desalinate the diluted SW, which stems from the final achievable FS recovery in FO. Instances of FO achieving 90 % and above FS recovery were observed under several operating conditions, but the realization of both 90 % FS recovery and RO energy below 1 kWh/m³ was confined to specific conditions. The optimal configuration was identified at an 80:20 feed-to-draw ratio with initial FS salinities of 0.25 and 0.5 g/l. In these cases, the recovered FS diluted DS down to 7.2 and 7.4 g/l salinity, enabling subsequent desalination with 0.95 and 0.96 kWh/m³ of RO SEC respectively. Under the 75:25 feed-to-draw volume ratio, the diluted DS ranged from 8.94 to 9.87 g/l (remaining below 10 g/l) within the considered initial FS salinity range, allowing subsequent desalination with RO energy consumption ranging from 1.02 to 1.07 kWh/m³. For feed-to-draw ratios of 50:50, 60:40, and 70:30, the SEC of RO in desalinating diluted DS, with salinity ranging from 10.67 to 18.25 g/l, exceeded 1 kWh/m³, ranging between 1.1 and 1.41 kWh/m³,

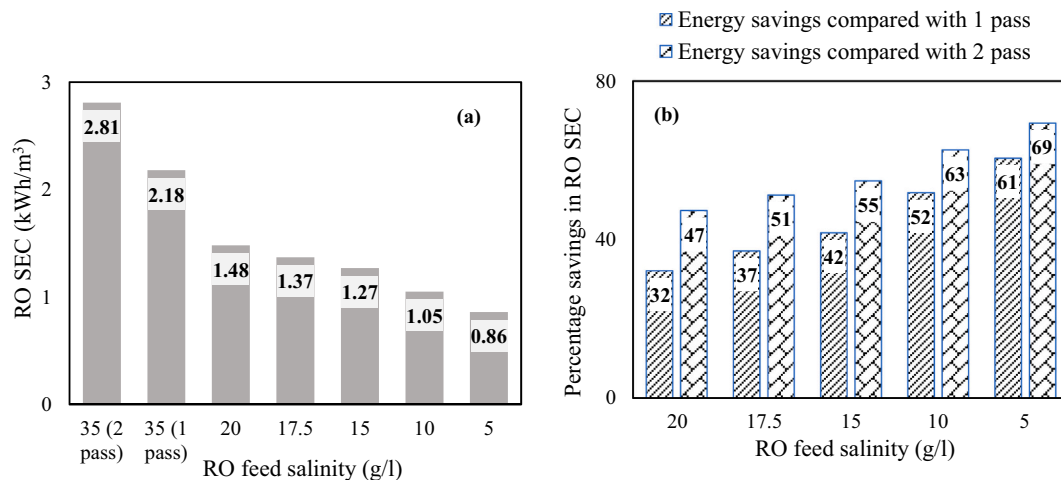


Fig. 7. (a) RO SEC for desalinating diluted feed to produce 1800 m³/h permeate at 45 % recovery. (b) Comparative energy savings with osmotically diluted RO feed vs. traditional one-pass and two-pass configurations.

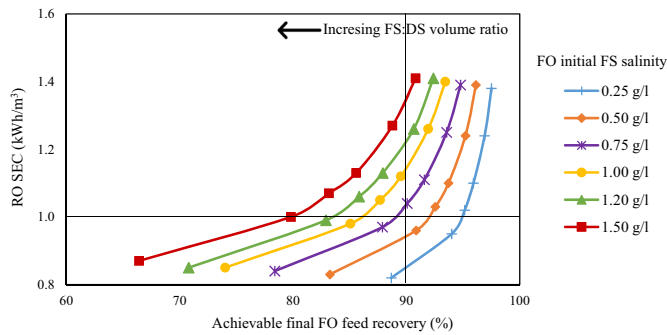


Fig. 8. SEC of RO in desalinating diluted SW corresponding to final achievable FS recovery under various operating conditions.

still lower than traditional RO system.

3.4. Enhanced permeate production

Demonstrating the compatibility with the transition to handle diluted feed, the existing plant with the same number of PVs and membrane elements, when subjected to enhanced recovery, showcased its ability to produce significantly higher permeate from diluted feed. It's noteworthy that the LewaPlus® software did not flag any warnings or errors during the simulation, even at these elevated recovery levels.

The RO system achieved recoveries well beyond the initial 45 % threshold, consistently surpassing 60 % with all feed salinities considered, as presented in Fig. 9. Specifically, with varying RO feed salinities of 20, 17.5, 15, 10, and 5 g/l, the plant achieved recoveries of 61 %, 63.5 %, 64.5 %, 67 %, and 67.5 %, respectively. Remarkably, with a feed salinity of 10 g/l, the plant exhibited an impressive 86 % increase in permeate production, while at 5 g/l, it reached 87.5 %. Operating at these levels, the SEC of RO stood at 1.65 and 1.36 kWh/m³, respectively. This SEC value was notably lower than that required for the desalination of full-strength SW, highlighting the positive impact of osmotic dilution on RO energy efficiency.

3.5. Considering new RO

Exploring the feasibility of considering a new RO facility to process the diluted feed for achieving high recovery shows great potential for improving energy efficiency in the desalination process (Fig. 10a). However, this improvement in energy savings was accompanied by the necessity of accommodating additional PVs ranging from 808 to 894 with each vessel containing 7 membrane elements, which presents certain cost considerations (Fig. 10b). The choice between retrofitting and constructing a new RO facility should be weighed carefully, considering the balance between energy efficiency and the associated investment in infrastructure.

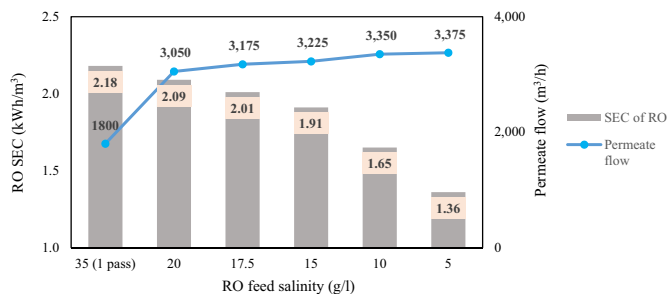


Fig. 9. SEC of RO for achieving enhanced recovery with various diluted SW feed.

3.6. Exploring energy demands of water reuse and desalination integration: FO-RO hybrid vs. independent schemes

The SEC of RO required to produce 1800 m³/h of permeate was compared for both cases. Within the FO-RO hybrid scenario, the optimized configuration, recovering 90 % from FO (with operating conditions: 80:20 feed-to-draw ratio, initial FS salinity of 0.5 g/l), was considered, and the resulting diluted SW with 7.42 g/l salinity served as the feed for RO stage. Similarly, in the case of independent schemes, 80 % of the total permeate produced originated from water reuse, while the remaining 20 % came from SWRO desalination (Fig. 11). The design parameters for this comparative study are outlined in Table 3.

SWRO desalination alone exhibited an elevated energy demand of 2.18 kWh/m³, aligning with the observed range in current state-of-the-art conventional SWRO plants [47] (Fig. 12). Contrastingly, water reuse alone accounted for a lower energy demand of 0.24 kWh/m³ as predicted by LewaPlus®, contradicting the reported values (~0.54 kWh/m³) in the literature [35,40]. Literature values were adopted for calculating energy demand in water reuse for independent schemes. Water reuse, even if accounting for the incorporation of MF/UF pre-treatment, requiring an extra specific energy input of 0.2 kWh/m³ [40] and additional energy consumption for the disinfection process 0.11 kWh/m³ [35,40] energetically outperforms SWRO desalination.

In independent schemes, the energy consumption of RO was 0.82 kWh/m³, with water reuse accounting for 0.43 kWh/m³ and SWRO for 0.39 kWh/m³, while for FO-RO hybrid, it was 0.96 kWh/m³. Energetically, independent schemes projected a 14.5 % advantage over FO-RO in terms of RO energy consumption. While FO-RO hybrid outperforms stand-alone SW desalination in terms of energy efficiency, the osmotic dilution benefits are not sufficient to surpass the advantages of independent schemes. Still, in an FO-RO hybrid system, WW reuse brings benefits beyond merely reducing RO energy demands. It may include the concentration of WW streams when implemented in the upstream of treatment scheme i.e., on the primary treated WW facilitating nutrients recovery and substantial cost savings resulting from bypassing energy intensive treatment steps [48], double dense membrane barrier to assure safe water reuse [49–51], generated brine is of lower concentration and can be discharged without adverse effects on the marine environment [22,27].

4. Conclusion

The FO simulation demonstrated the feasibility of achieving a robust 90 % WW recovery in many instances under diverse operating conditions. However, realizing both 90 % WW recovery from FO and maintaining the subsequent RO desalination SEC below 1 kWh/m³ was possible only under specific conditions. The optimal configuration that facilitated achieving both objectives was identified at an 80:20 feed-to-draw volume ratio and an initial FS salinity of 0.5 g/l in FO. This allowed SW to dilute to 7.4 g/l, enabling subsequent RO desalination with 0.96 kWh/m³, showcasing the feasibility of existing desalination plants to process diluted SW. Furthermore, resulting from osmotic dilution of SW, existing plant with the same design achieved elevated recovery rates of up to 67 %.

It is evident that water reuse alone has a lower energy consumption compared to all treatment techniques. However, when considering combining water reuse and desalination, despite independent schemes being 14.5 % more energy-efficient than FO-RO hybrid, the latter still holds potential. The FO-RO hybrid, beyond reducing RO SEC, shows potential in concentrating WW streams for valuable nutrients recovery, serving as a double-dense barrier for safe water reuse, with the generated RO brine of lower concentration suitable for safe discharge. Examining fouling phenomena in the FO process and assessing model performance and operation in counter-current mode would provide valuable insights for future studies. Additionally, studying the economics of membrane costs and lifespan variations under different

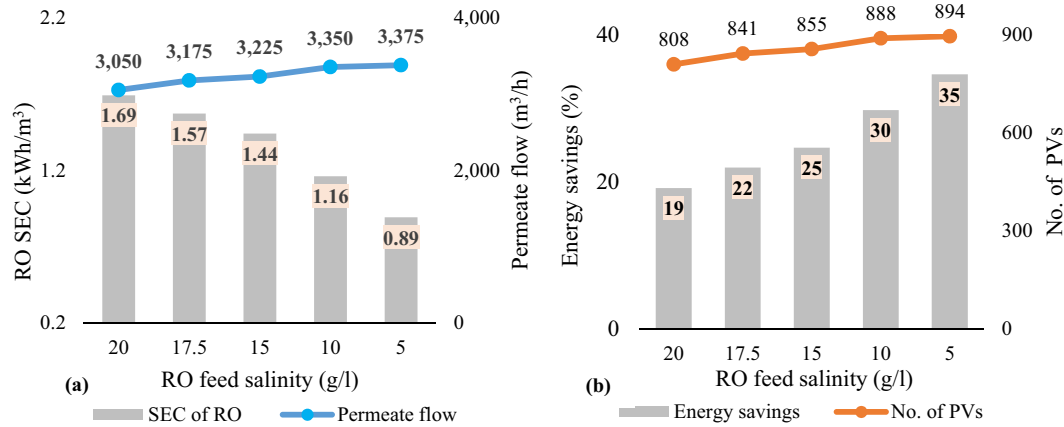


Fig. 10. (a) SEC of new RO for achieving enhanced recovery with various diluted SW feed. (b) Energy savings in comparison with existing RO and corresponding no. of PVs required w.r.t considering new RO.

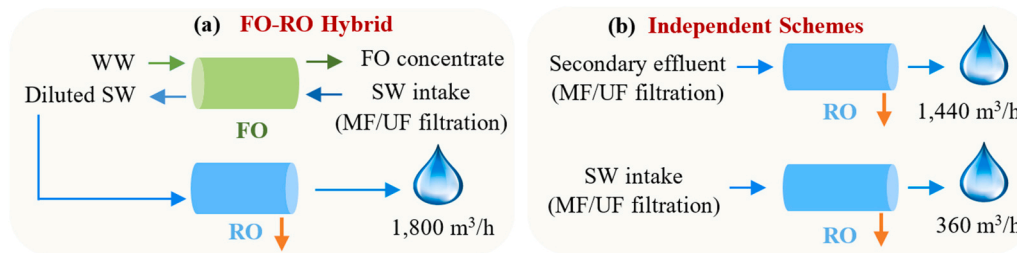


Fig. 11. Distribution of permeate production in (a) FO-RO hybrid and (b) independent schemes. MF = micro-filtration, UF = ultra-filtration.

Table 3

Design parameters for comparative study employed in LewaPlus®.

Treatment technique	RO feed salinity (g/l)	R (%)	RO permeate flow (m³/h)	Membrane element	Stage 1 (No. of PVs, No. of Els/PV)	Stage 2 (No. of PVs, No. of Els/PV)
SWRO (1 pass)	35	45	1800	SW HR 400 34	(477,7)	–
Water reuse	0.5	80	1,800	BW LP 400 34 SD	(230,7)	(116,7)
FO-RO	7.42	45	1800	SW HR 400 34	(477,7)	–
Independent schemes	0.5	80	1440	BW LP 400 34 SD	(184,7)	(93,7)
SWRO (1 pass)	35	45	360	SW HR 400 34	(96,7)	–

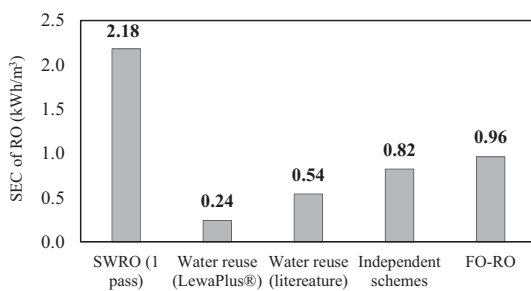


Fig. 12. Comparative SEC of RO in various water treatment techniques.

operating conditions could offer further understanding and optimization strategies for practical implementation.

CRedit authorship contribution statement

Rajashree Yalamanchili: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ignasi Rodriguez-Roda:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Albert Galizia:** Writing – review & editing, Software, Resources. **Gaëtan Blandin:** Writing – review & editing, Visualization,

Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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